

ChessAIThon

Strategic Gaming Meets AI in VET Education

Agenda

1. **Chess Representation** in Computers
2. **Data Management** & Storage
3. **Chess AI Architecture**: ChessMarro
4. **Training & Datasets**
5. **Performance Results**
6. **Human-AI Collaboration**

Chess Representation

- **Human:** FEN, PGN, SAN, UCI
- **Computer:** Bitboards & Piece-centric structures
- **ChessNet:** **77x8x8 Bitboard** multi-channel input
- Encodes positions, turns, and **legal moves**

Data & Storage

- **Standards:** PGN (Games) & FEN (Positions)
- **Modern Pipeline:** CSV, JSON, and **Parquet**
- **Parquet:** Efficient compression for 77-channel tensors
- **Version Control:** Git/GitHub for dataset collaboration

AI Architecture

- Inspired by **AlphaZero** & Leela Chess Zero
- **ChessNet**: Convolutional Neural Network (CNN)
- **MCTS**: Monte Carlo Tree Search for lookahead
- Combines **Policy** (Move) and **Value** (Win Chance)

Training Strategy

- **CNN:** 256+ channel Residual Towers
- **Dataset:** Human & simulated Stockfish data
- **Fine-tuning:** Adapts to specific endgame scenarios
- **Self-Reinforcement:** Future potential for AlphaZero-style play

The Dataset

- **Scale:** ~1.5 Million scenarios
- **Origin:** Lichess games & Stockfish 400-node evals
- **Format:** 77x8x8 Parquet tensors
- **Optimization:** Transformed via Kaggle & Colab GPU nodes

Performance Results

- **Rank 1 Accuracy:** ~28% match with Stockfish
- **Top-5 Accuracy:** ~70% (Top best moves)
- **Insight:** High Top-K accuracy provides strong priors for MCTS
- Plays in a more "**human-like**" style than Stockfish

Decision Logic

- **Simulation:** MCTS explores CNN's best policies
- **Noise:** Dirichlet Noise prevents predictability
- **Refinement:** Simulation corrects raw CNN "intuition" errors
- **Stochastic:** Softmax decision making for variety

Community AI

- **ChessMinds:** Web platform for crowdsourced data
- **Open Source:** Libraries shared on GitHub & HuggingFace
- **Participatory:** Students train, measure, and refine models
- **Web App:** Deployable with CI/CD integration

Thank You

Questions & Collaboration

GitHub: [xxjcaxx/ChessAIThon](https://github.com/xxjcaxx/ChessAIThon)

HuggingFace: [jocasal/chessmarrov1](https://huggingface.co/jocasal/chessmarrov1)